

CS-UY 2214 — Recitation 8

Introduction

Complete the following exercises. Unless otherwise specified, put your answers in a plain text file named `recitation8.txt`. Number your solution to each question. When you finish, submit your file on Gradescope. Then, in order to receive credit, you must ask your TA to check your work. Your work should be completed and checked during the recitation session.

You may consult the E20 manual, which is available on Brightspace.

Problems

1. Consider the multicycle E20 processor, as described in the E20 manual. Name all of the micro-operation registers.

2. Consider the multicycle E20 processor, as described in the E20 manual.

Which control signals would you set to achieve each of the following microcode instructions?

For each microcode instruction, give all relevant control signals and their values. Each instruction is to be taken independently.

- (a) `pc <- pc + 1`
- (b) `R[7] <- MDR`
- (c) `M[pc] <- B`

3. Consider the multicycle E20 processor. For each of the following opcodes, tell which states are necessary to execute it. Recall that each instruction uses exactly five states. Give each state by its number as indicated on the E20 multicycle state diagram.

For example, `add` uses states 0, 1, 4, 7, and 14.

- (a) `or`
- (b) `lw`
- (c) `addi`
- (d) `jal`

4. Consider the multicycle E20 processor. Consult the circuit diagram in the E20 manual. Using only that diagram and the instruction reference, determine which E20 instruction will be performed with the following control line settings for each of the stages:

```
IF stage:
    WEir = 1
    WEram = 0
    Wereg = 0
    WEpc = 0
```

```
WEa = 0
WEb = 0
WEalu = 0
MUXaddr = 1
ID stage:
WEa = 1
WEb = 1
MUXr1 = 4
MUXb = 0
EXEC stage:
aluOp = 0
WEalu = 1
MUXalu = 1
MEM stage:
MUXaddr = 0
WEram = 0
WB stage:
WEpc = 1
MUXpc = 0
```